Concentration Units

Answer key with worked solutions

A. Ideas

1. Molarity and molality.

> (a) **Molarity** is moles of solute per **litre of solution**; **molality** is moles of solute per **kilogram of solvent**. (b) **Molarity** changes with temperature, because the volume of a solution expands or contracts. Molality, mass percent, ppm and mole fraction use **masses** and amounts, which do not change.

B. Mass-based units

2. Sugar.

> Mass of solution $= 12.0\ \text{g} + 188\ \text{g} = 200.\ \text{g}$; $\dfrac{12.0\ \text{g}}{200.\ \text{g}} \times 100\% = \mathbf{6.00\%}$

3. Fluoride.

> 0.45 mg = 0.00045 g; $\dfrac{0.00045\ \text{g}}{500.\ \text{g}} \times 10^{6} = \mathbf{0.90\ ppm}$

4. Saline.

> $0.0090 \times 500.\ \text{g} = \mathbf{4.5\ g\ NaCl}$

C. Moles-based units

5. Glucose.

> Moles: glucose $\dfrac{18.0\ \text{g}}{180.16\ \text{g/mol}} = 0.09991\ \text{mol}$; water $\dfrac{90.0\ \text{g}}{18.02\ \text{g/mol}} = 4.994\ \text{mol}$
>
> (a) $x = \dfrac{0.09991\ \text{mol}}{0.09991\ \text{mol} + 4.994\ \text{mol}} = \mathbf{0.0196}$
>
> (b) $m = \dfrac{0.09991\ \text{mol}}{0.0900\ \text{kg}} = \mathbf{1.11\ mol/kg}$

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D. Converting between units

6. Sulfuric acid.

> Take 1.000 L: mass $= 1000\ \text{mL} \times 1.84\ \text{g/mL} = 1840\ \text{g}$
>
> Mass of $H_2SO_4 = 0.980 \times 1840\ \text{g} = 1803\ \text{g}$; moles $= \dfrac{1803\ \text{g}}{98.08\ \text{g/mol}} = 18.38\ \text{mol}$
>
> $c = \dfrac{18.38\ \text{mol}}{1.000\ \text{L}} = \mathbf{18.4\ mol/L}$

7. From molality to mass percent.

> Take 1.000 kg of water: it contains $1.50\ \text{mol NaCl} = 1.50\ \text{mol} \times 58.44\ \text{g/mol} = 87.66\ \text{g}$.
>
> $$\%\ \text{NaCl} = \frac{87.66\ \text{g}}{1000\ \text{g} + 87.66\ \text{g}} \times 100\%$$
> $$= \mathbf{8.06\%}$$

E. Challenge

8. From molality to molarity.

> Take 1.000 kg of water: $3.00\ \text{mol NaCl} = 3.00\ \text{mol} \times 58.44\ \text{g/mol} = 175.3\ \text{g}$.
>
> Mass of solution $= 1000\ \text{g} + 175.3\ \text{g} = 1175.3\ \text{g}$
>
> Volume of solution $= \dfrac{1175.3\ \text{g}}{1.12\ \text{g/mL}} = 1049\ \text{mL} = 1.049\ \text{L}$
>
> $c = \dfrac{3.00\ \text{mol}}{1.049\ \text{L}} = \mathbf{2.86\ mol/L}$

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